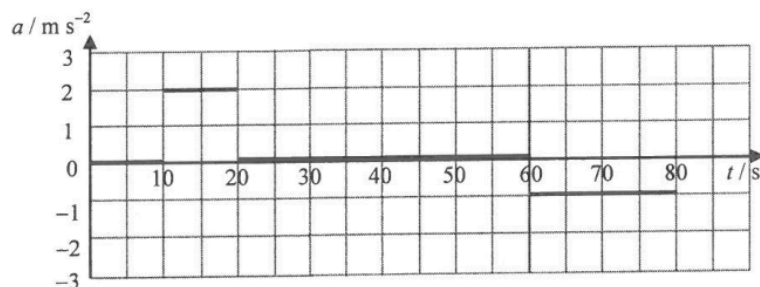


3. (a) 0 – 10 s: constant / uniform acceleration  
10 – 80 s: constant velocity / uniform motion

|    |
|----|
| 1A |
| 1A |
| 2  |
| 1A |
| 1A |
| 2  |

- (b) (i) Car B.  
It is represented by the steepest slope between 10 s and 20 s for B.  
 $a = \frac{20 - 0}{20 - 10} = 2 \text{ m s}^{-2}$

(ii)



Uniform acceleration between 10 s and 20 s & uniform deceleration between 60 s and 80 s.

Or Either acceleration or deceleration correct.

All correct.

|    |
|----|
| 1A |
| 1A |
| 1A |
| 2  |

- (c) (i) Total area of A by 20 s =  $\frac{(10 + 20)}{2} \times 15 = 225 \text{ m}$   
Total area of B by 20 s =  $\frac{(10 \times 20)}{2} = 100 \text{ m}$   
Separation of A & B,  $s_{AB} = 225 \text{ m} - 100 \text{ m} = 125 \text{ m}$

|    |
|----|
| 1M |
| 1A |
| 2  |

- (ii) Let B meet A at time  $(20 + T) \text{ s}$ ,  
 $(v_B - v_A) \times T = s_{AB}$   
 $(20 \text{ m s}^{-1} - 15 \text{ m s}^{-1}) \times T = 125 \text{ m}$   
 $T = 25 \text{ s}$

$\therefore$  B catches up with A at  $t = (20 + 25) \text{ s} = 45 \text{ s}$ .

Or  $s_A + v_A \times T = s_B + v_B \times T$   
 $225 \text{ m} + 15 \text{ m s}^{-1} \times T = 100 \text{ m} + 20 \text{ m s}^{-1} \times T$   
Or Accept using graphical method and read the answer from the graph

|    |
|----|
| 1A |
| 1M |
| 1M |
| 1A |
| 2  |

- (d) Let  $F_A$  and  $F_B$  be the driving forces developed by the engines of A and B respectively.  
Ratio of power developed by cars A and B,  
 $P_A : P_B = F_A \times v_A : F_B \times v_B$   
 $= v_A^2 \times v_A : v_B^2 \times v_B$   
 $= 3^3 : 4^3 = 27 : 64$

|    |
|----|
| 1M |
| 1A |
| 2  |

4. (a) The block decelerates uniformly (until at rest or its velocity is zero at  $t = 1.5$  s).  
It then moves with uniform acceleration down the plane (until  $t = 3.5$  s).

1A

1A

2

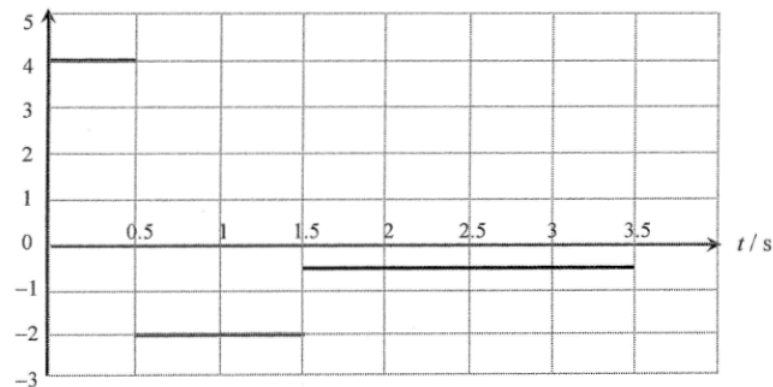
(b) (i)  $a_2 = \frac{-1 - 0}{3.5 - 1.5}$   
 $= 0.5 \text{ m s}^{-2}$

1M

1A

2

(ii)

 $a / \text{m s}^{-2}$ 

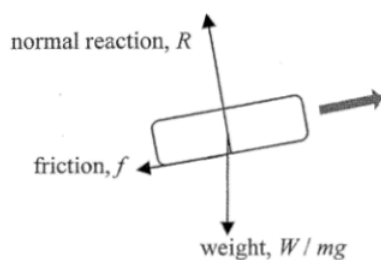
Either acceleration correct  
All correct

1A

1A

2

(c)



Friction correctly indicated  
All correct

1A

1A

2

(d) Moving upward :  $-mg \sin \theta - f = ma$   
 $-(1)(9.81) \sin \theta - f = (1)(-2) \dots \dots \dots \textcircled{1}$

1M

Moving downward :  $-mg \sin \theta + f = ma'$   
 $-(1)(9.81) \sin \theta + f = (1)(-0.5) \dots \dots \dots \textcircled{2}$

1M

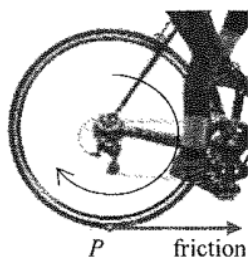
$\textcircled{2} - \textcircled{1} : 2f = 1.5$   
 $f = 0.75 \text{ N}$   
(Note:  $\theta = 7.32^\circ$ )

1A

3

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                         |                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------|
| 3. (a) $a = \frac{6-0}{2-0}$<br>$= 3 \text{ m s}^{-2}$ (downwards)                                                                                                                                                                                                                                                                                                                                                                                                              | 1M<br>1A                                |                                                                                         |
| (b) A: 395 N      B: 569 N      C: 685 N<br><br>In stage B, balance reading = weight (Newton's 1 <sup>st</sup> law),<br>$mg = m \times 9.81 \text{ m s}^{-2} = 569 \text{ N}$<br>$m = 58.0 \text{ kg}$<br><br><div style="border: 1px solid black; padding: 5px; display: inline-block;">             Or In stage A, according to Newton's 2<sup>nd</sup> law<br/> <math>(569 - 395) \text{ N} = ma = m(3 \text{ m s}^{-2})</math><br/> <math>m = 58.0 \text{ kg}</math> </div> | 2<br>1A<br><br>1M<br>1A<br><br>1M<br>1A | For $g = 10 \text{ m s}^{-2}$ , $m = 56.9 \text{ kg}$                                   |
| (c) (i) For stage C, by Newton's 2 <sup>nd</sup> law,<br>$F = ma$<br>$(569 - 685) \text{ N} = (58.0 \text{ kg}) a$<br>$a = -2 \text{ m s}^{-2}$<br><br>Thus, $a = \frac{0-6}{T-12} = -2 \text{ m s}^{-2}$<br>$T = 15 \text{ s}$                                                                                                                                                                                                                                                 | 3<br><br>1M<br><br>1M                   | For $g = 10 \text{ m s}^{-2}$ , $m = 56.9 \text{ kg}$ ,<br>$a = -2.04 \text{ m s}^{-2}$ |
| (ii) Height $\approx$ displacement of lift = area under graph<br>$= \frac{(12-2)+15}{2} \times 6$<br>$= 75 \text{ m}$                                                                                                                                                                                                                                                                                                                                                           | 2<br>1M<br><br>1A<br><br>2              |                                                                                         |

3. (a)



1A

1

- (b) Mechanical power delivered (for work done against the resistive force)

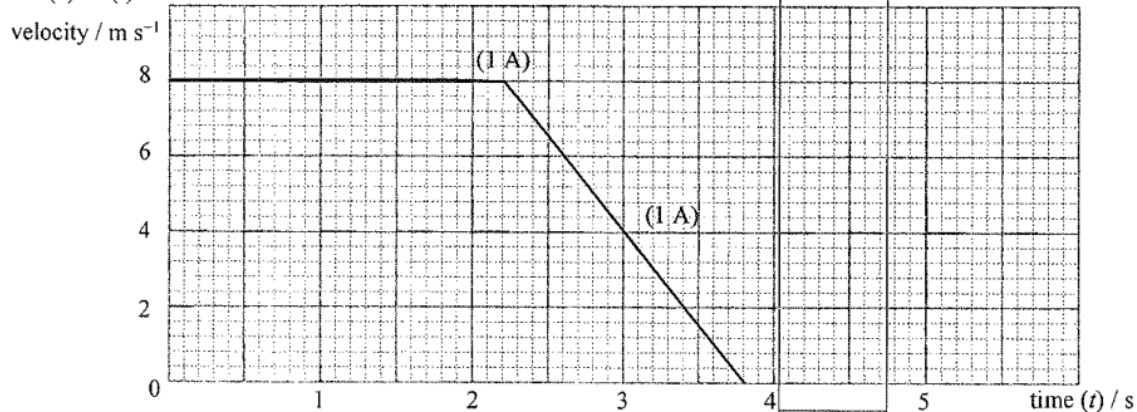
$$\begin{aligned}
 P &= Fv \\
 &= 17.0 \times 8.0 \\
 &= 136 \text{ W}
 \end{aligned}$$

1M

1A

2

- (c) (i)



2

- (ii) Deceleration:  $\left| \frac{0-8}{3.8-2.2} \right| = 5.0 \text{ m s}^{-2}$   
 Resistive force =  $65 \times 5.0$   
 $= 325 \text{ N}$

1M

e.c.f. from (c)(i)

1A

*Alternative method:*

$$\text{Braking distance} = \frac{1}{2}(8)(3.8 - 2.2) = 6.4 \text{ m}$$

$$Fs = \frac{1}{2}mv^2 - 0$$

$$F(6.4) = \frac{1}{2}(65)(8)^2$$

$$F = 325 \text{ N}$$

1M

1A

2

- (iii) Total distance travelled after  $t = 2.0 \text{ s}$   
 $=$  area under the graph from  $t = 2.0 \text{ s}$  to  $t = 3.8 \text{ s}$   
 $= \frac{0.2+1.8}{2} \times 8.0 = 8.0 \text{ m}$   
 Distance between the bicycle and the obstacle when the bicycle stops =  $9 - 8$   
 $= 1 \text{ m}$

1M/1A

1M

1A

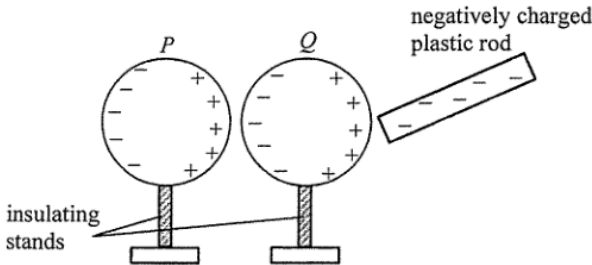
3

- (d) The padded cushion can increase the contact time of the knee and the ground such that a smaller force acts on the knee.

1A

1A

2

|                                                                                                                                                                |                                                                                   |                                                                           |   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------|---|
| 9. (a)                                                                                                                                                         |  | 2A                                                                        | 2 |
| (b) No, it remains unchanged as charging is through induction (without any contact or transfer of charges).                                                    | 1A+1A                                                                             | 2                                                                         |   |
| (c) Hold / bring the unknown charged rod (assume positive) near sphere P, which carries positive charges, without touching it.                                 | 1A                                                                                | 2                                                                         |   |
| If P is repelled by the rod, the rod should carry positive charges.<br><u>And / Or</u><br>If P is attracted by the rod, the rod should carry negative charges. | 1M                                                                                | 1M for the correct procedure to identify the polarity of the charged rod. |   |

3. (a) The object is moving to the left with constant deceleration and (momentarily) stops at  $t = 4$  s.

(b) Displacement = total area of the graph

$$= \frac{1}{2} \times (-10) \times 4 + \frac{1}{2} \times 15 \times (10 - 4)$$

$$= 25 \text{ m}$$

Position of the object is 25 m from point  $O$  on the right side.

(c) Net force acting on the object

$$= ma$$

$$= 0.4 \times \frac{15 - (-10)}{10 - 0}$$

$$= 1 \text{ N (towards right)}$$

|    |   |
|----|---|
| 1A | 2 |
| 1M | 2 |
| 1A | 2 |
| 1M | 2 |
| 1A | 2 |

Or

$$v^2 = u^2 + 2as$$

$$15^2 = (-10)^2 + 2 \times \frac{15 - (-10)}{10 - 0} \times s$$

$$s = 25 \text{ m}$$